

Career Guidance Agent

Personalised Career Recommendation System

- React + Vite
- Supabase Auth
- 7 Faculties
- 28+ Career Paths
- PDF Reports

7

Faculties

28+

Career Paths

100+

Certifications

6

Questions

A faculty-aware career recommendation system that uses real datasets to guide students toward the right career paths, certifications, short courses, and job-readiness actions — powered by Supabase.

SECTION 01 · PROJECT OVERVIEW

About the Career Guidance Agent

The Career Guidance Agent is a full-stack web application that provides personalised career path recommendations to university students and recent graduates across all seven faculties. By combining a structured self-assessment quiz with a real-dataset scoring engine, it helps users discover which career paths best align with their academic background, skills, and professional goals.

Built with React 18 + Vite on the frontend and Supabase for authentication and data persistence, the system is fast, accessible, and mobile-friendly. Reports are automatically saved to a cloud database and can be retrieved and exported as PDF.

Project Name	Career Guidance Agent
Version	1.0.0
Target Users	University students & recent graduates (all 7 faculties)
Platform	Web application — React 18 + Vite 4
Backend	Supabase (PostgreSQL 17 + Auth)
Scoring Engine	Weighted algorithm + Real-data mode (5,000-record Kaggle dataset)
Salary Context	South African Rands (R) — localised SA market ranges
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SECTION 02 · PROBLEM STATEMENT

The Challenge This System Solves

University students and recent graduates across all faculties frequently face significant uncertainty when transitioning from academic study into the professional world. Despite completing years of structured education, many graduates lack clear, personalised guidance on which specific career paths are most aligned with their skills, strengths, and goals.

Lack of Personalisation

Generic career advice does not account for individual strengths, academic background, or personal goals. One-size-fits-all resources fail to reflect the diversity of student profiles.

Information Overload

The internet contains vast amounts of career information, but students struggle to filter what is relevant to their specific faculty, skills, and South African context.

No Actionable Roadmap

Even when career paths are identified, students often do not know which certifications to pursue, which short courses to take, or what skills to develop to become job-ready.

Report Loss on Refresh

Existing lightweight tools (static quizzes, PDFs) do not persist results, meaning students lose their career assessment whenever they close or refresh the browser.

Salary Uncertainty

Students lack access to localised, realistic salary expectations in South African Rands for each career path, making financial planning difficult.

Proposed Solution

The Career Guidance Agent addresses all of the above by providing a structured, faculty-aware self-assessment quiz that produces a personalised career roadmap — complete with certifications, short courses, key skills, salary ranges in Rands, and role-specific action steps. Reports are saved automatically to Supabase and can be retrieved and exported as a formatted PDF at any time.

SECTION 03 · SYSTEM ARCHITECTURE

How the Application Works

The diagram below illustrates the end-to-end flow of the Career Guidance Agent, from user authentication through career scoring to report generation and PDF export.

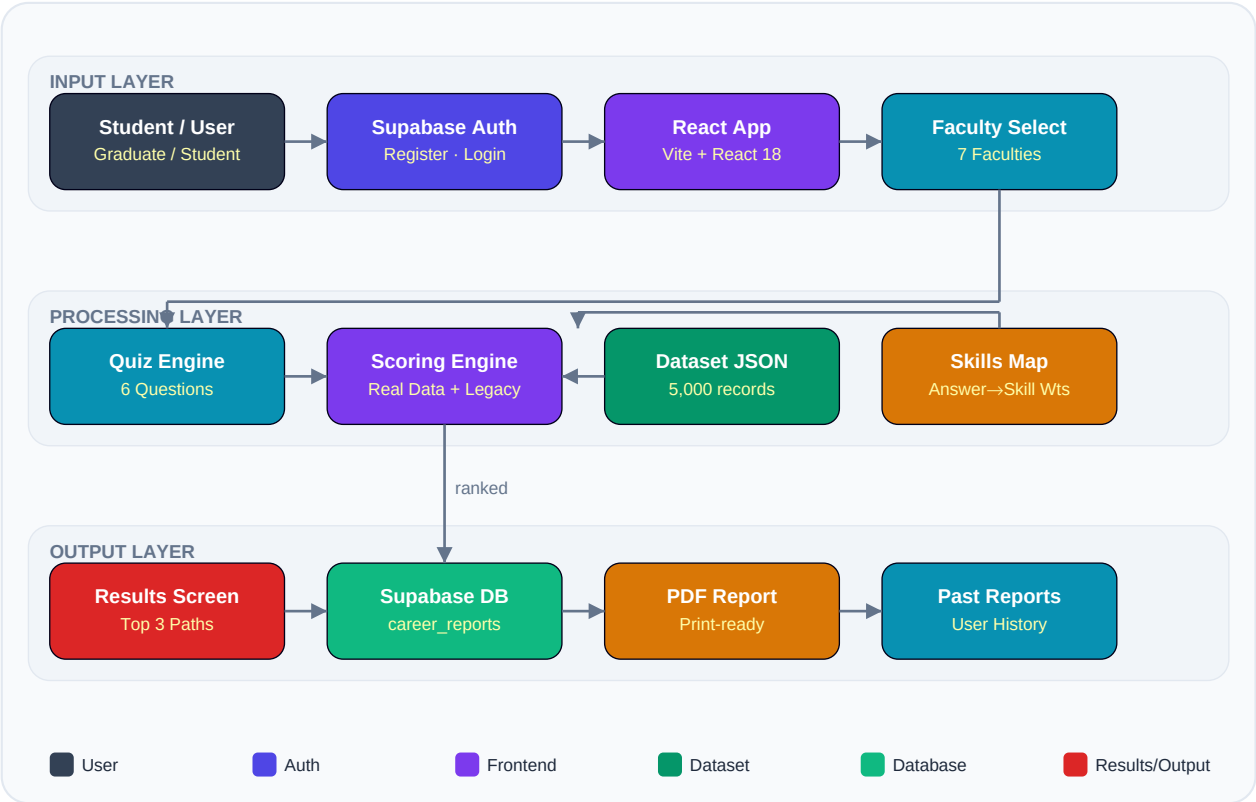


Figure 1 — End-to-End System Architecture

Architecture Layers

Layer	Component	Responsibility
Authentication	Supabase Auth	Email/password register & login. Session persists across refreshes.
Frontend	React 18 + Vite	SPA. Screens: Auth → Welcome → Faculty → Quiz → Results.
Data Layer	faculties.js / questions.js	7 faculties, 28+ career paths, 6 quiz questions, scoring weights.
Scoring Engine	scorePaths()	Blends real-dataset skill frequencies (70%) with legacy weights (30%).
Real Data	career_data.json	Processed from 5,000-record Kaggle dataset via process_dataset.py.
Skills Map	skillsMap.js	Translates quiz answers into dataset skill names and weights.
Database	Supabase PostgreSQL	Stores career reports: user, faculty, assessment, results, date.
PDF Export	Browser Print API	HTML report opened in new tab; user saves as PDF via browser.

SECTION 04 · FEATURES & TECHNOLOGY

What the System Does and How It Is Built

Core Features

User Authentication

Secure email/password registration and login via Supabase Auth. Sessions persist across refreshes. Logout redirects to the Register page.

Faculty-Aware Assessment

Students select one of 7 university faculties (Arts & Design, Economics & Finance, Engineering & Built Environment, Humanities, ICT, Management Sciences, Science). Career paths are filtered and scored for the selected faculty.

6-Question Quiz

Universal questions covering: preferred activity, personal strength, desired impact, comfort with numbers, career timeline, and primary career goal. Answers feed the scoring engine.

Dual Scoring Engine

When career_data.json is present (from the 5,000-record Kaggle dataset), real skill-frequency data powers recommendations. Otherwise, calibrated legacy weights are used.

Personalised Career Roadmap

Top 3 matching career paths shown as expandable cards — each with certifications, short courses, key skills, salary ranges (R), and role-specific job-readiness steps.

Cloud Report Persistence

Every completed assessment is automatically saved to Supabase PostgreSQL. Reports survive page refresh and browser close.

PDF Export

Past reports downloadable as fully formatted, print-ready PDFs showing all career path details and the universal job-readiness checklist.

Scoring Mode Badge

Results screen shows a green "Real-Data Mode" badge when powered by the Kaggle dataset, or grey "Built-in Scoring" when using legacy weights.

Technology Stack

Category	Technology	Purpose
Frontend	React 18	Component-based UI with hooks and state management
Build Tool	Vite 4	

Styling	Custom CSS	Faculty-themed UI with CSS variables (--fac-color)
Auth & Database	Supabase	PostgreSQL 17 + Auth + REST API
DB Client	@supabase/supabase-js v2.45	Frontend Supabase SDK
Data Processing	Python 3 + pandas + openpyxl	Dataset processor: process_dataset.py
PDF Generation	Browser Print API	HTML to PDF via window.print()
Dataset	Kaggle (5,000 records)	career_dataset_large.xlsx

SECTION 05 · USER JOURNEY

Step-by-Step Flow Through the Application

1 — Register or Log In	The user lands on the authentication screen and creates an account (name, email, password) or logs in. Supabase manages session tokens. App shows a loading indicator while the session resolves on page refresh.
2 — Welcome Screen	Personalised with the user's first name. If previous reports exist, a "My Reports (N)" button appears to view and download them as PDF.
3 — Faculty Selection	The user selects one of 7 faculties. Each card is colour-coded and the selection updates the app's entire colour theme via the --fac-color CSS variable.
4 — Quiz (6 Questions)	The user answers 6 questions covering preferred activity, personal strength, desired impact, comfort with numbers, career timeline, and primary career goal. A progress bar tracks completion; the Back button allows corrections.
5 — Results & Auto-Save	The top 3 career paths are scored and displayed as expandable cards. The report is auto-saved to Supabase. A status banner confirms the save. The header shows which scoring mode was used (Real-Data or Built-in).
6 — PDF Export	From the Welcome screen's "My Reports" panel, any past report can be downloaded as a formatted PDF showing: top career matches with full details, certifications, courses, skills, salary ranges, role actions, and the job-readiness checklist.

SECTION 06 · DATABASE SCHEMA

Supabase PostgreSQL Structure

The application uses a single Supabase PostgreSQL table to store career reports. Row Level Security (RLS) is enabled with open insert and select policies.

Column	Type	Description
id	uuid (PK)	Auto-generated primary key
user_name	text NOT NULL	Display name from auth user metadata
user_email	text	Email from authenticated Supabase user
faculty	text NOT NULL	Selected faculty name (e.g. "ICT")
assessment	jsonb NOT NULL	Quiz answers keyed by question ID
report	jsonb NOT NULL	Top 3 career paths with full details
created_at	timestampz	Auto-set timestamp (DEFAULT now())

Supabase Project Details

Project Name	CareerguidanceAgent
Project ID	visjzvscdxrpkrbkaibo
Region	eu-west-1 (Europe — West Ireland)
Database	PostgreSQL 17
Auth	Email / Password (Supabase Auth)
RLS Policies	Allow insert for all · Allow select by email

File Structure

CareerGuidanceAgent/	
career-guidance-agent.html	Standalone HTML version (all 7 faculties)
showcase.html	Landing page / live demo interface
process_dataset.py	Dataset processor: Excel -> JSON
career-guidance-react/	
src/	
App.jsx	Root component, auth, data loading
components/	
AuthScreen.jsx	Register / Login UI
NavBar.jsx	Navigation + user avatar + logout
WelcomeScreen.jsx	Hero + past reports panel
FacultyScreen.jsx	7 faculty selection cards
QuizScreen.jsx	6-question animated quiz
ResultsScreen.jsx	Ranked paths + auto-save + mode badge

data/	
faculties.js	7 faculties, 28+ paths, scoring engine
questions.js	6 quiz questions + readiness checklist
skillsMap.js	Quiz answer -> dataset skill mapping
lib/	
supabase.js	Supabase client + auth + report helpers
public/	
career_data.json	Generated by process_dataset.py
start.bat	Windows launcher script

SECTION 07 · SETUP & DEPLOYMENT

Running the Application

Prerequisites

- Node.js v18+ (nodejs.org)
- Python 3.8+ (for dataset processing — optional)
- Supabase account with the CareerguidanceAgent project active
- Kaggle dataset: career_dataset_large.xlsx (optional — enables Real-Data Mode)

Quick Start

1. Open a terminal inside the career-guidance-react\ folder.
2. Run start.bat (Windows) — installs dependencies and starts the dev server.
3. Open <http://localhost:5173> in your browser.
4. Register an account, select a faculty, complete the quiz, view results.

Enabling Real-Data Mode (optional)

1. Download career_dataset_large.xlsx from Kaggle (link in EDA notebook).
2. Place it in CareerGuidanceAgent\ (same folder as process_dataset.py).
3. Run: `pip install pandas openpyxl` then `python process_dataset.py`
4. Script writes career-guidance-react/public/career_data.json automatically.
5. Restart dev server — Results screen shows green "Real-Data Mode" badge.

Supabase Notes

- Free-tier Supabase projects pause after 7 days of inactivity. Restore via dashboard.
- Disable email confirmation: Dashboard -> Authentication -> Providers -> Email -> turn off Confirm email.
- The project uses the publishable (anon) key — never put the secret key in frontend code.

SECTION 08 · FUTURE ENHANCEMENTS

Planned Improvements

ML-Based Scoring

Train a classification model (Random Forest / XGBoost) on the 5,000-record dataset and serve predictions via a Supabase Edge Function for fully data-driven recommendations.

Live Job Market Data

Pull live job listings from LinkedIn / Careers24 API to supplement recommendations with current SA market demand and actual vacancy counts per career path.

Admin Dashboard

Build a faculty-admin view to monitor student assessment trends, popular career paths, faculty distribution, and report generation statistics.

Mobile App

Wrap the React app in a React Native shell for iOS/Android distribution to reach students who primarily use mobile devices.

Email Report Delivery

Send completed PDF reports directly to the user's registered email via Supabase Edge Functions and the Resend API — no manual download required.

SA-Localised Dataset

Expand the scoring engine with South African job market data — pulling CGPA equivalents, SA-specific certifications (ECSA, SABPP, HPCSA), and local employer preferences from SETA and DHET reports.

Career Guidance Agent

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Supabase Project: visjzvscdxrpkrbkaibo · Region: eu-west-1
Dataset: Kaggle — ahsanneural/career-path-recommendations-dataset
